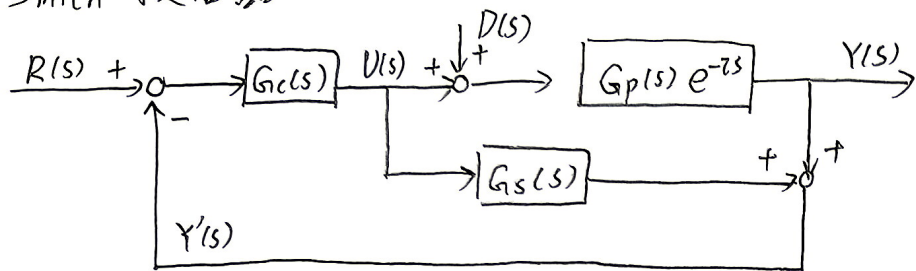


Smith 预估器

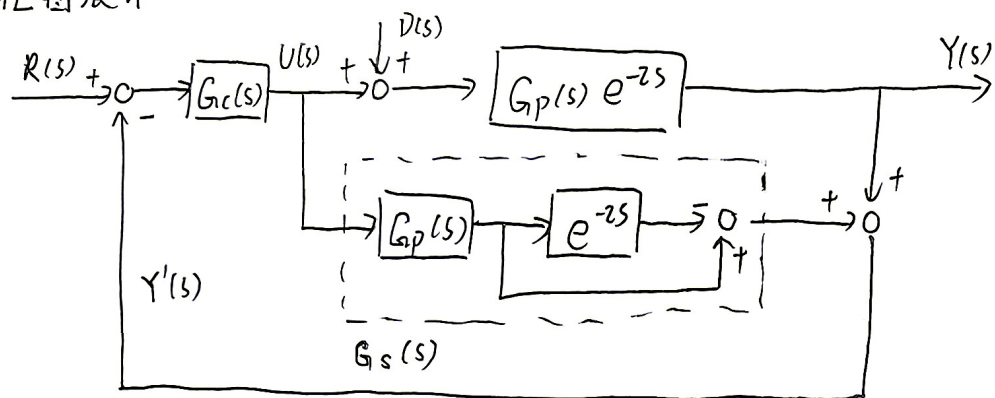


$$G(s) = G_p(1 - e^{-zs})$$

改善延迟

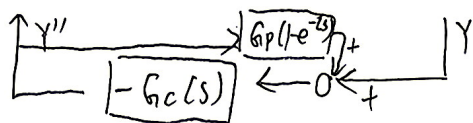
以模型为基础的预估器补偿方法

框图展开

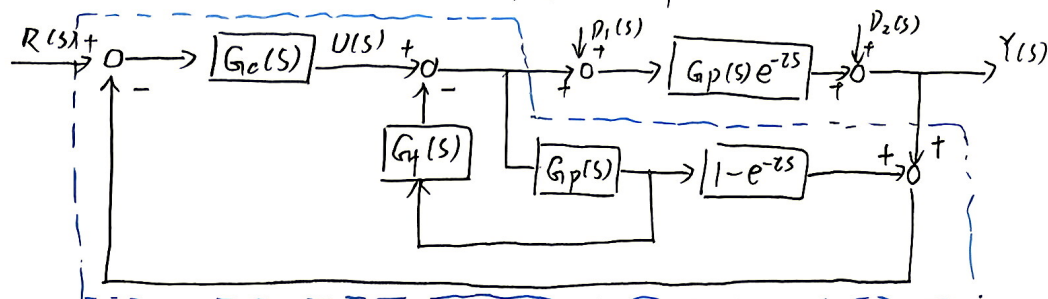


$$\begin{cases} Y = G_c G_p e^{-zs} (R - Y') \\ Y' = \frac{Y}{G_p e^{-zs}} + G_p (1 - e^{-zs}) + Y \end{cases} \Rightarrow \frac{Y(s)}{R(s)} = \frac{G_c(s) G_p(s) e^{-zs}}{1 + G_c(s) G_p(s)}$$

$$\begin{cases} Y = G_p(s) e^{-zs} (D + Y'') \\ Y'' = \frac{-G_c(s)}{1 + G_c(s) G_p(s) (1 - e^{-zs})} Y \end{cases} \Rightarrow \frac{Y(s)}{D(s)} = \frac{[1 + G_c(s) G_p(s) (1 - e^{-zs})] G_p e^{-zs}}{1 + G_c(s) G_p(s)}$$

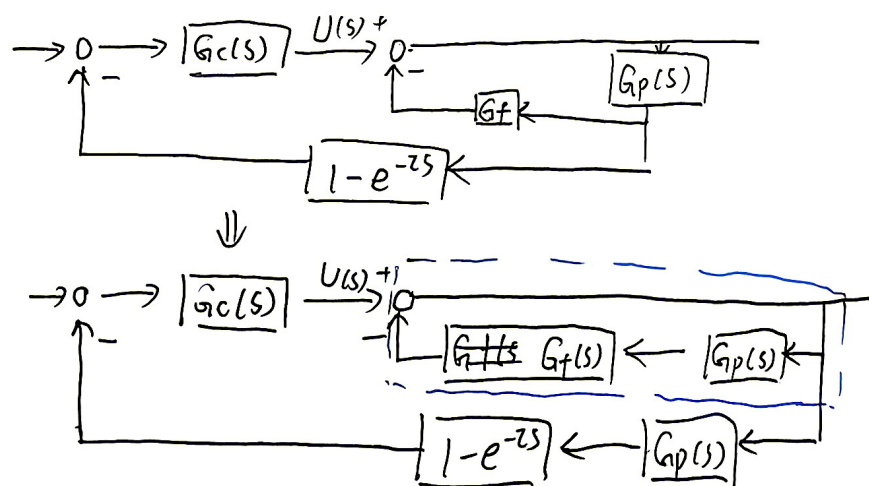


完全抗干扰的 Smith 预估器可行性分析



$$G_f(s) = \frac{G_c(s) G_p(s) (e^{-zs} - 1)}{G_p(s)} \quad (\text{推导: 对 } D_1 \text{ 或 } D_2 \text{ 扰动})$$

只看框中部分



$$\begin{aligned} \text{框中内回路传递函数 } G_{11}(s) &= \frac{1}{1 + G_f G_p} = \frac{1}{1 - [1 + G_c(s) G_p(s) (e^{-zs} - 1)]} \\ &= \frac{1}{-G_c G_p (1 - e^{-zs})} \end{aligned}$$

$$\text{子系统传递: } G_{12}(s) = \frac{G_c G_{11}}{1 + G_c G_{11} G_p (1 - e^{-zs})} = \frac{G_c G_{11}}{0} \rightarrow \infty$$

子系统失去了稳定性, 整个系统失稳, 方案不可行